

AI Artifact Audit Log

This section documents AI interactions. Entries continue the cumulative log from D1-D5 (entries #01-25), targeting ≥ 25 entries total by the end of W10.

Entry #01 - W1 / P1	
Field	Content
DTC Component	Problem & Business Decomposition
AI Tool / Prompt	Claude – "We are developing an IS project for Information System Analysis and Design on Agricultural UAVs in the Mekong Delta. Suggest 3 distinct sub-domain directions, stating the primary business problem and implementation complexity for each."
AI Output (tóm tắt)	3 directions: Fleet Management Generic tracking; Precision Spraying Platform localized scheduling; Marketplace hourly drone rentals. All rated 3 stars complexity.
Decision & Refinement	SELECTED: Precision Spraying Platform: Empirical data: Nguyen et al.2025, N=940. Pain points validated: HTX uses Zalo/Excel. The platform layer is vacant in the ecosystem. REJECTED: Scope too broad, no IS depth. REJECTED: Overlaps with another team's confirmed topic.
Verification	Verification: Ecosystem gap confirmed - Lao Dong (2025) confirms digital scheduling problem remains unresolved in ĐBSCL cooperatives.

Entry #02 - W1 / P1	
Field	Content
DTC Component	Pattern Recognition & Comparison
AI Tool / Prompt	ChatGPT – "Compare Agricultural UAV, Delivery UAV, and Surveillance UAV across: stakeholder complexity, regulatory burden in Vietnam, and data availability for an IS academic project."
AI Output (tóm tắt)	AI concluded Delivery UAV is 'easiest' due to lower perceived regulation. Agricultural UAV ranked medium complexity.
Decision & Refinement	AI RANKING REJECTED incorrect evaluation: Agricultural UAV has a more structured legal framework (Decree 288/2025/ND-CP + Plant Protection Dept) than Delivery UAV. Maintained Agricultural UAV: (a) peer-reviewed Research available (Nguyen 2025), (b) verifiable field incidents, (c) richer stakeholder map for IS analysis.
Verification	Verification: Decree 288/2025/ND-CP cross-checked via Thuvienphapluat confirms specific drone ID and pilot certification logging requirements.

Entry #03 - W1 / P1	
Field	Content
DTC	Decomposition - Ecosystem Mapping

Component	
AI Tool / Prompt	Claude – "For the AgriUAV Platform (ITA301, Mekong Delta): decompose the current Agricultural UAV ecosystem into structural layers. Identify which layers are occupied and which represent market gaps for an IS solution."
AI Output (tóm tắt)	AI proposed a 2-layer split: Hardware/Device layer and Application/User layer. Suggested the gap lies in 'application integration.' No distinct Service Layer or Platform Layer identified.
Decision & Refinement	REJECTED 2-layer model oversimplified and inaccurate. Team defined 3-Layer Model: Hardware Layer (DJI/XAG occupied), Service Layer (HTX fleets occupied, manual), Platform Layer (digital IS coordination VACANT).
Verification	Verification: 3-layer model validated by Thanh Nien (June 2025) : 1,215 active drones across 3 provinces with no platform-layer management software reported.

Entry #04 - W2 / P1	
Field	Content
DTC Component	Decomposition - Stakeholder Mapping
AI Tool / Prompt	Claude – "Project: AgriUAV Platform for drone spraying in Mekong Delta. List all stakeholders, classify as Primary/Secondary/Regulatory, and outline core roles and expectations for each."
AI Output (tóm tắt)	Suggested 6 general groups: Farmers, HTX, Pilots, Drone Manufacturers (DJI), Software Developers, and Local Government totaling ~12 stakeholders.
Decision & Refinement	5 core stakeholders accepted. Removed 'Software Developer Team' - this is the project team, not an external stakeholder. SUPPLEMENTED 2 critical actors AI omitted: Operations Dept./General Staff (sole authority for UAV flight permits under Decree 288/2025); Pesticide Retailers (essential for inventory/compliance integration). Split generic 'Government' into 2 specific regulatory bodies: Plant Protection Dept. + Operations Dept.
Verification	Verification: Operations Dept. authority confirmed via Vietnam Naval News (2024).

Entry #05 - W2 / P1	
Field	Content
DTC Component	Decomposition - AS-IS Process Analysis
AI Tool / Prompt	Claude – "Decompose the AS-IS business process of an agricultural cooperative in Vietnam executing drone spraying services. List each step, the actor responsible, and

	pain points at each stage. Context: currently managed via Zalo/Excel."
AI Output (tóm tắt)	Proposed a basic 4-step generic workflow: Intake → Dispatch → Execution → Reporting. High level pain points only: lack of tracking, slow report compilation.
Decision & Refinement	Accepted 4-step structure as baseline. Expanded to 5 steps added 'Monitoring & Control' as a distinct phase (AI merged this into Execution). Localized pain points: Phase 3 operators may under-dose chemicals or inflate flight hours (AI missed this risk entirely); Phase 4 Long An 110kV power line incident - [CORRECTION: nguồn chính xác là Tuổi Trẻ (2024), không phải Dân Việt 2026. Đã loại khỏi tất cả bản nộp cuối (D1 v1.6+).] as concrete geofencing evidence; Phase 5 linked compliance failure directly to Decree 288/2025/NĐ-CP.
Verification	Verification: 110kV incident: Tuổi Trẻ (2024) drone collision in Long An Province, 76,000 customer blackout.

Entry #06 - W2 / P1	
Field	Content
DTC Component	Process / Algorithm - Business Case
AI Tool / Prompt	Claude – "Provide an industry-standard business case structure for an IS research project, including a preliminary ROI template and a formal problem statement formulation suitable for a university-level IS analysis project."
AI Output (tóm tắt)	Standard 5-part structure (Context → Problem → Impact → Solution → Scope) and a generic theoretical ROI formula: (cost saved / investment) × 100%.
Decision & Refinement	Adopted structural blueprint as foundation. Applied Mekong Delta data localization: 15–20% [CORRECTION: con số 15–20% chi phí vận hành trong AI output ban đầu được AI trích dẫn nhầm từ "Nguyen et al. 2025". Sau khi team kiểm tra, không tìm thấy nguồn peer-reviewed nào chứa con số này. Quyết định: relabel thành "team estimate derived from published operational benchmarks" - ghi rõ trong D1 v1.6 Entry #06 và Section 2.3 methodology note.] operational cost figure from Nguyen et al. (2025) N=940; 1,215 drones across 3 provinces from Thanh Nien June 2025. Added HTX Cost Benefit table (not in AI output): pesticide -30%, fuel -20%, labor -75% = ~21.9M VND savings vs. 3.6M VND platform cost → positive ROI from Season 1.
Verification	Verification: Nguyen et al. (2025). Province drone counts: Thanh Nien (June 2025) .

Entry #07 - W3 / P1	
Field	Content
DTC Component	Pattern Recognition - Business Case Comparison
AI Tool / Prompt	Claude – "Compare the business case of an Agricultural UAV platform vs. a Delivery UAV platform in Vietnam: market size, competition level, scalability, and suitability for IS research."
AI Output (tóm tắt)	AI concluded Delivery UAV has 'a larger market' because logistics is booming. Agricultural UAV was rated lower on 'tech sophistication.'

Decision & Refinement	COUNTER ARGUMENT AGAINST AI BIAS: AI conflated 'market size = revenue potential' with 'research quality.' ITA301 evaluates domain richness, not revenue. Agricultural UAV is more suitable: (1) peer-reviewed paper exists (Nguyen et al. 2025, RWAE Journal) - no equivalent for Delivery UAV in Vietnam; (2) real field data available by province from Vietnamese press; (3) richer stakeholder map; (4) 'Platform layer vacant' claim provable via 3-layer ecosystem decomposition. Team maintained Agricultural UAV selection. AI bias pattern recognized and documented.
Verification	Verification: Pattern recognition - team identified AI bias toward 'market = revenue' framing vs. ITA301 standard of 'domain richness = research quality.'

Entry #08 - W3 / P1	
Field	Content
DTC Component	Decomposition - WBS & Risk Register
AI Tool / Prompt	Claude – "Propose a Risk Register for a 10-week IT project analyzing agricultural drone spraying in ĐBSCL. List 8–10 risks with likelihood (H/M/L), impact, and mitigation strategy."
AI Output (tóm tắt)	AI listed 10 risks: scope creep, timeline, team conflict, data quality, tool learning curve, stakeholder availability, budget, integration complexity, security breach, deployment failure.
Decision & Refinement	Accepted 5 generic risks (scope creep, timeline, team conflict, stakeholder availability, data quality). REMOVED 'deployment failure' - ITA301 does not require actual deployment. REMOVED 'budget' - not applicable to a university project. SUPPLEMENTED 4 UAV-domain-specific risks AI missed: Regulation change risk (NĐ 288/2025 newly effective, supplementary decrees possible); Domain knowledge gap (team lacks direct field experience - probability set to Medium, corrected from Low); Literature availability; Stakeholder interview difficulty.
Verification	Verification: NĐ 288/2025 confirmed effective via VnBusiness and Government E-Portal.

Entry #09 - W4 / P2 Requirements	
Field	Content
DTC Component	Process / Algorithm
AI Tool / Prompt	Claude – "Design a requirement elicitation workflow for an Agricultural UAV project: steps from stakeholder identification to SRS finalization, including techniques for each step (interview, observation, prototyping...)."
AI Output (tóm tắt)	Proposed a 5-step workflow: Identify stakeholders → Elicit requirements → Analyze → Specify → Validate. Techniques: structured interview, JAD session, use case workshop.
Decision & Refinement	Accepted the workflow structure. Added 'Domain immersion' step: read Cuc BTV reports and Nguyen 2025 BEFORE interviews. Added 'Roleplay interview': team acts as

	farmer/HTX manager since direct stakeholder access is limited. Note: AI did not suggest role-play for students with no field access.
Verification	Role-play interview confirmed in W4 instructor guidance. Domain immersion cross-checked with NFR Availability.

Entry #10 - W4 / P2 Requirements	
Field	Content
DTC Component	Decomposition
AI Tool / Prompt	Claude – "Propose Functional Requirements for AgriUAV Platform, modules: Field Mapping, Spray Scheduling, Fleet Dispatch, Real-time Monitoring. Each FR has ID, description, priority (Must/Should/Could)."
AI Output (tóm tắt)	Listed 18 FRs classified by 4 modules. Covered GPS mapping, schedule creation, drone assignment, telemetry display.
Decision & Refinement	Accepted 14 core FRs. Removed 2 FRs: 'AI pest detection' and 'autonomous flight path optimization' - out of IS scope (AI/robotics). Merged 2 FRs about 'drone status display' into FR10. Added 2 regulatory FRs AI missed: FR13: Store drone ID + pilot cert per Decree 288 (FR-REG01); FR14: Export reports in Plant Protection Dept. format (FR-REG02).
Verification	FR-REG01/02 verified directly against Decree 288/2025/NĐ-CP text and Cục Trồng trọt & BVTV requirements .

Entry #11 - W4 / P2 Requirements	
Field	Content
DTC Component	Process / Algorithm
AI Tool / Prompt	Claude – "Write Non-Functional Requirements for Agricultural UAV Platform with quantitative thresholds: Performance (drone position update), Reliability (uptime during season), Usability (limited IT farmers), Scalability, Security."
AI Output (tóm tắt)	Suggested NFRs: update ≤ 5 seconds, uptime 95%, usability 10 min/task, 100 drones concurrent, AES-256 encryption.
Decision & Refinement	Accepted NFR structure calibrated for ĐBSCL context: • Performance: $\leq 5s \rightarrow \leq 3s$ (drone moves $\sim 5-7$ m/s; 5s too slow for safety alerts) • Reliability: 95% $\rightarrow$ 99% (peak season 30-60 days; downtime unacceptable) • Usability: 10 min $\rightarrow \leq 5$ min (Nguyen 2025: avg age ~ 51, limited IT) • Scalability: kept 50-drone reference workload per HTX (AI suggested 100, beyond

	evaluation baseline) Added NFR06 Availability: offline mode when no 4G - AI omitted entirely.
Verification	3s threshold based on DJI Agras T40 actual speed. Age ref: Nguyen 2025 N=940. Offline req cross-checked with Mekong Delta field conditions. Drone phun thuốc trừ sâu va vào dây 110kV gây mất điện ở 5 huyện - Tuổi Trẻ Online Drone: "Trợ thủ" mới của nhà nông

Entry #12 - W4 / P2 Requirements	
Field	Content
DTC Component	Decomposition
AI Tool / Prompt	Claude – "List ≥8 Use Cases for AgriUAV Platform with actor, precondition, main flow (3–5 steps), alternative flow, postcondition. Focus on spray scheduling, drone dispatch, real-time monitoring, alerts."
AI Output (tóm tắt)	Generated 10 use cases with full main flow but only 2/10 UCs had alternative flows. UC about regulatory reporting was missing entirely.
Decision & Refinement	Accepted 9 UCs. Removed UC from 'User Registration' - too simple, not a standalone UC. Added missing alt flows for 6 UCs: • UC03: Alt flows for GPS outside 100m radius and offline check-in (no 4G) • UC04: Alt flow when 4G connection lost mid-session • UC05: Alt flow when operator does not respond in 5 min • UC06: Alt flow when weather changes while drone is in-flight (manager override + audit trail) • UC07: Alt flow when actual pesticide usage deviates >20% from plan (reason required) • UC08: Alt flow when a pilot certificate expired during the reporting period (compliance note) Added UC10: View Spraying History & Export VietGAP - AI missed regulatory traceability UC entirely.
Verification	Alt flow 'offline 4G' cross-checked with NFR06. UC10 from FR15 (VietGAP) and Decree 288/2025 audit trail requirements. Nghị định 288/2025/NĐ-CP quản lý tàu bay không người lái và phương tiện bay khác mới nhất Toa đàm "Đưa drone trở thành trợ thủ đắc lực của nhà nông" - CropLife Việt Nam

Entry #13 - W4 / P2 Requirements	
Field	Content
DTC Component	Pattern Recognition / Consistency & Counter-argument
AI Tool / Prompt	Claude – "Review D2 v2.4.3 against the D1 v1.5.1 proposal and the RBL workshop criteria; identify scope/actor/telemetry/traceability/validation inconsistencies and propose concrete corrections."
AI Output (tóm tắt)	Identified 15 issues across 4 severity levels: telemetry semantics mismatch between Scope (mobile-GPS proxy / simulated) and FR10–FR12/UC04–UC06 (described drone position, battery, pesticide as real sensor data); UC10 used "Export Company" as a system actor outside the 5-role RBAC; UC03 Alt 3a allowed Pilot self-bypass of the 100m GPS check; UC05 battery/pesticide data source undefined; traceability statement claimed "no orphaned FR" despite cross-cutting FR19/20; NFR05 mixed requirement

	with technology choice; NFR04 mixed 1-HTX baseline with 10-HTX planning; legal retention and report format lacked specific article references; UC07 introduced biometric/PIN as a hidden requirement; FR06 rule set narrower than UC02.
Decision & Refinement	Accepted the diagnostic; applied surgical corrections rather than rewriting: (a) Telemetry: clarified data source in FR10, FR11 and UC05 (mobile-GPS proxy or structured simulated telemetry within MVP); MAVLink remains post-MVP per Section 2.5. (b) UC10 actor: changed Primary Actor to HTX Manager acting on behalf of the contracted Export Company, consistent with the 5-role RBAC (FR18) and D1's post-MVP exporter integration. (c) UC03 Alt 3a: removed Pilot self-bypass; HTX Manager approval required with audit-trail entry. (d) UC07: replaced fingerprint/PIN with authenticated-session confirmation already covered by FR18, eliminating the hidden requirement. (e) FR06: added battery, pilot certificate validity, and pilot schedule to the rule set. (f) NFR04: separated MVP acceptance baseline (1 HTX, ~20 req/s) from a future-cluster planning assumption (10 HTX, ~200 req/s). (g) NFR05: rephrased as solution-neutral security outcomes with current minimum baselines. (h) Retention 5y and report format relabelled as working assumptions pending authority confirmation. (i) Traceability: dropped "no orphaned" phrasing; introduced explicit Coverage Type classification (Direct / Indirect / Cross-cutting). (j) Added Section 2.2 fix removing "automatic confirmation" (inconsistent with UC01 Pending-confirmation flow). (k) Section 1.1 "signed agreement" softened to "baseline working agreement, pending stakeholder sign-off".
Verification	Each correction cross-checked against D1 v1.5.1 (canonical) and the RBL workshop checklist. Counts confirmed after edits: 21 FR, 7 NFR, 10 UC, 8 US, 5 RBAC roles - all preserved.

Entry #14 - W5 / P3 Modeling	
Field	Content
DTC Component	Abstraction
AI Tool / Prompt	Claude – "Đề xuất external entities, data flows và sub-processes cho DFD Level 1 của AgriUAV Platform."
AI Output (tóm tắt)	AI đề xuất 6 sub-processes cho DFD Level 1: (1) Field Mgmt, (2) Scheduling, (3) Fleet Dispatch, (4) Monitoring, (5) Notification, (6) User Management. 8 external entities.
Decision & Refinement	Chấp nhận 4 sub-processes cốt lõi: Field Mgmt, Scheduling, Monitoring, Compliance. Loại Notification và User Mgmt - đây là cross-cutting services, không phải standalone process ở L1 theo Gane-Sarson convention. AI không phân biệt được cross-cutting vs standalone process - Human Delta quan trọng.
Verification	Cross-check DFD balanced rule - tất cả data flow vào/ra Level 0 phải xuất hiện đúng ở Level 1.

Entry #15 - W6 / P3 D3 Class	
Field	Content
DTC Component	Abstraction
AI Tool / Prompt	Claude – "Vẽ Class Diagram cho AgriUAV Platform với đầy đủ attributes và methods."
AI Output (tóm tắt)	AI sinh Drone class với 25+ attributes (firmware_version, sensor_list, obstacle_avoidance_level...) và 18 methods bao gồm cả low-level hardware ops.
Decision & Refinement	Chấp nhận 9 core attributes cho Drone: droneId, serialNumber, model, tankCapacityL, batteryPercent, gpsLat, gpsLng, status, cooperativeld. Loại 16 attributes còn lại - thuộc phạm vi drone manufacturer SDK, không thuộc IS database. Gộp 5 methods phức tạp thành 3 core: isAvailable(), updatePosition(), sendAlert().
Verification	Kiểm tra với FR06 (dispatch cần status), FR10 (monitoring cần gpsLat/Lng), FR11 (alert cần batteryPercent) - 9 attributes đủ để fulfill tất cả FR liên quan.

Entry #16 - W6 / P3 D3 Seq	
Field	Content
DTC Component	Process / Algorithm
AI Tool / Prompt	Claude – "Tạo Sequence Diagram cho UC03 - On-site GPS Check-in."
AI Output (tóm tắt)	AI sinh sequence cho UC03 với 5 messages (happy path only). Không có alternative flows, không có offline scenario.
Decision & Refinement	Chấp nhận happy path cơ bản. Bổ sung ALT 3a: GPS outside 100m radius → manual override với reason (FR09 business rule). Bổ sung ALT 4a: Offline mode (no 4G) → local cache → auto-sync ≤30s khi reconnect (NFR06). v1.4: chuẩn hóa thuật ngữ cache offline thành IndexedDB.
Verification	Cross-check với D2 SRS NFR06 (offline ≤30s sync, IndexedDB) và UC03 Alt flows.

Entry #17 - W6 / P3 D3 Activity	
Field	Content
DTC Component	Process / Algorithm
AI Tool / Prompt	Claude – "Thiết kế Activity Diagram cho quy trình monitoring drone đang bay, 3 loại alert, escalation timers, parallel processing."
AI Output (tóm tắt)	AI đề xuất activity diagram tuyến tính (sequential) cho AD2: check geofence → check battery → check weather → update map. Không có fork/join, escalation timers, repeat loop, swim lanes.
Decision & Refinement	PHẢN BIỆN: Sequential diagram sai về mặt kỹ thuật - geofence alert có thể bị delay khi đang wait weather API response (drone bay 5–7 m/s, 1s = 5–7m drift). Thiết kế lại với fork/join 3 checks đồng thời, escalation timers (5 phút UC05, 10 phút UC06, force-suspend 30 phút), repeat loop, swim lanes.
Verification	Cross-check với UC05/UC06 escalation trong D2 SRS. Tham chiếu DJI Agras T40: tốc độ bay tối đa 7 m/s.

Entry #18 - W8 / P4 Architecture

Field	Content
DTC Component	Pattern Recognition
AI Tool / Prompt	Claude – "So sánh Layered vs Microservices vs Event-Driven cho AgriUAV HTX ĐBSCL ~50 drones."
AI Output (tóm tắt)	AI kết luận Microservices 'tốt nhất cho scalability' với reasoning generic (Netflix/Uber style). Event-Driven xếp hạng 2.
Decision & Refinement	AI bias toward 'trending architecture' - Microservices overhead không phù hợp scope HTX ĐBSCL (1 cooperative = 50 drones = 5 modules; HTX không có DevOps team; inter-service latency ~50–100ms không phù hợp NFR01 $\leq 3s$; team 5 SV/10 tuần). Final choice: Layered Architecture.
Verification	NFR01 ($\leq 3s$), NFR04 (50-drone reference workload), NFR07 (maintainability) - Layered thỏa mãn tất cả.

Entry #19 - W9 / P4 Security	
Field	Content
DTC Component	Process / Algorithm
AI Tool / Prompt	Claude – "STRIDE threat model AgriUAV: airspace data, PII nông dân, telemetry, compliance logs."
AI Output (tóm tắt)	AI đề xuất 6 STRIDE threats generic (Spoofing, Tampering, Repudiation, Info Disclosure, DoS, EoP) với mitigation generic (JWT, HTTPS, logging, rate limiting).
Decision & Refinement	STRIDE framework. Tinh chỉnh từng threat: • Spoofing → 'Fake GPS check-in' (100m radius + manualOverride flag) • Tampering → 'WebSocket MITM inject false telemetry' (TLS 1.3 + message signing) • Repudiation → append-only FlightLog + DB audit trigger • Information Disclosure → gpsTrail AES-256 + RBAC field-level • EoP → JWT role tampering (server-side validation)
Verification	OWASP Top 10, NFR05 (TLS 1.3, AES-256, JWT), Decree 288/2025/NĐ-CP append-only audit requirement (working assumption).

Entry #20 - W6 (extended) / P3→P4 Transition	
Field	Content
DTC Component	Pattern Recognition / Consistency Audit
AI Tool / Prompt	Claude – "Audit D3 v1.4 against D1 v1.6 and D2 v2.4; identify structural/consistency gaps; produce updated D3 v1.5 core content while leaving UML/DFD diagram sections as placeholders."
AI Output (tóm tắt)	Claude rà soát D3 v1.4 đối chiếu với D1 v1.6 và D2 v2.4. Phát hiện 5 nhóm vấn đề đồng bộ: (1) Use Case Diagram chưa tồn tại như một artifact riêng; (2) DFD Level 1 thiếu rationale khi loại Operations Dept; (3) Class Diagram chưa có method cho FR21 và confirmDosage() cho FR02; (4) ngôn ngữ "GEOFENCE VIOLATION" mâu thuẫn với "information-level alert" của FR12; (5) Sequence Diagram UC03 Alt 3a vẫn cho phép Pilot tự xác nhận override GPS.
Decision & Refinement	Tạo Use Case Diagram mới (UseCaseDiagram_v1.5.puml) - đóng gap (1). Cập nhật DFD (level0_v1.5, level1_v1.5): loại Operations Dept có rationale; khôi phục n1–n5; Export Company → "post-MVP" - đóng gap (2). Cập nhật Class Diagram (v1.5): thêm registerDrone()/updateDroneStatus() (FR21), confirmDosage() (FR02), requestCheckInOverride() - đóng gap (3) và một phần gap (5). Chuẩn hóa "VIOLATION"

	→ "information-level alert" trong core content - đóng một phần gap (4). Sequence UC03 CHƯA được vẽ lại (theo yêu cầu Minh - UML để trống) - ghi thành outstanding item.
Verification	Đối chiếu trực tiếp với D1 v1.6, D2 v2.4 (FR01–FR21, UC01–UC10, NFR04/NFR05 hedge), ClassDiagram_v1.5.puml, UseCaseDiagram_v1.5.puml, DFD_Agruav_level0_v1.5.drawio, DFD_Agruav_level1_v1.5.drawio.

Entry #21 - W7 / P4 Design	
Field	Content
DTC Component	Process / Algorithm - Compliance Enforcement
AI Tool / Prompt	Claude - designed the audit log schema and evaluated whether application-layer enforcement was sufficient or whether database-layer enforcement was required.
AI Output (tóm tắt)	The AI initially proposed using generic system_logs, with actor_id configured as ON DELETE SET NULL. No database-level constraints were implemented to prevent UPDATE or DELETE operations, relying solely on application-level conventions.
Decision & Refinement	The proposal was rejected because enforcing compliance only at the application layer does not satisfy the immutable audit trail requirement under Decree No. 288/2025/NĐ-CP. Application-layer validation can be bypassed through direct database writes or implementation bugs. The table was renamed to audit_logs, and two database triggers (prevent_audit_log_modification and trg_audit_logs_append_only) were introduced to block UPDATE and DELETE operations at the database engine level. In addition, the foreign key actor_id was changed from ON DELETE SET NULL to ON DELETE RESTRICT to prevent deletion of users with existing audit records. User deactivation is handled through the is_active flag instead of physical deletion.
Verification	Verified against Design Specification 3 (which defines the append-only audit log convention), Section 7.1 (STRIDE – Repudiation), and NFR05.

Entry #22 - W7 / P4 Design	
Field	Content
DTC Component	Critical Thinking & Trade-off
AI Tool / Prompt	Gemini – "Evaluate and propose the system architecture (Layered Architecture vs. Microservices) based on the 200 req/s load requirement of the AgriUAV project."
AI Output (tóm tắt)	AI analyzed the trade-off matrix, indicating that Microservices have an absolute advantage in Scalability, but warned about high infrastructure operational costs and deployment complexity.
Decision & Refinement	Decided to adopt Layered Architecture. Argued that the actual context at Mekong Delta cooperatives involves low budgets and a lack of dedicated DevOps teams. Microservices would violate the NFR07 commitment (hotfix resolution time ≤4 hours). Layered Architecture is fully sufficient to meet the MVP requirements.
Verification	Cross-checked against practical barriers (Cost) and NFR07 (Maintainability). Aligns with the RBL Guide's Justification section.

Entry #23 - W7 / P4 Design	
Field	Content

DTC Component	Traceability & Consistency
AI Tool / Prompt	Gemini – "Cross-check data consistency between the Physical Database Schema design draft (D4 v0.51) and the System Models document (D3 v4.2)."
AI Output (tóm tắt)	AI detected a critical inconsistency: D3 v4.2 utilizes UUID v4, whereas the D4 draft's SQL DDL code uses INT AUTO_INCREMENT for primary keys.
Decision & Refinement	Firmly rejected the current D4 draft and refactored the entire DDL code to strictly use the UUID v4 standard (VARCHAR(36)). UUID is mandatory for resolving distributed data synchronization issues when drones operate in network dead zones (Offline Mode), preventing primary key collisions upon network reconnection.
Verification	Traceability ensures 100% alignment with D3 v4.2 and directly fulfills NFR06 (Availability / Offline Sync).

Entry #24 - W7 / P4 Design	
Field	Content
DTC Component	Critical Thinking & Trade-off
AI Tool / Prompt	Gemini – "Design the data flow for the Telemetry module (UC04) to continuously ingest flight coordinates transmitted from drone devices to the server."
AI Output (tóm tắt)	AI proposed having the API Gateway receive the coordinate data and directly write it to the PostgreSQL/TimescaleDB database to ensure real-time tracking.
Decision & Refinement	Rejected direct database writes. Analyzed that at an operating frequency of 4Hz across 50 drones, the system must process 200 requests/sec, which would cause severe disk I/O bottlenecks. Inserted a Redis In-Memory Buffer layer as a "shock absorber" to cache high-frequency telemetry, utilizing background workers for scheduled asynchronous batch inserts into the long-term database.
Verification	Verified against NFR04 (Scalability threshold: 50 drones/HTX) and ensures compliance with data retention policies under Decree 288/2025/NĐ-CP.

Entry #25 - W9 / P4 Design Data Sync	
Field	Content
DTC Component	Process / Algorithm - Data Sync
AI Tool / Prompt	Claude – "Design the data synchronization mechanism for the pilot's app when transitioning from offline field operations back to online status."
AI Output (tóm tắt)	Proposed a simple database dump that uploads the entire local SQLite dataset onto the server upon detecting a network connection. Suggested using standard auto-incrementing integer keys (INT AUTO_INCREMENT) for primary database records to save storage space.
Decision & Refinement	Human Delta: Firmly rejected integer keys. Since the platform operates heavily on an Offline-First model where records are generated on the field without server connection, auto-increment keys would cause massive primary key collisions during sync. Enforced UUID v4 across all entity definitions.
Verification	Global Database Data Dictionary in D4.

